M. Sc. Physics (Semester-II) 2026
Quantum Mechanics II – Tutorial 1

1. An one dimensional harmonic oscillator has a potential $V(x) = \frac{1}{2}m\omega^2 x^2$. Also consider the turning points appearing at $x = \pm c$. Find the estimate for the energy of the bound state using WKB. Also, check that under which condition there will be only one bound state within the turning point region (need to show a relation among the parameters).

2. **Bouncing ball problem:** A ball of mass m is moving under the gravitational potential

$$\begin{aligned} V(x) &= mgx \quad x > 0 \\ &= 0 \qquad x < 0, \end{aligned}$$

   where x stands for the height from the hard surface and g is gravitational constant. Find the quantized energy levels of the bouncing ball problem, assuming $E > 0$.

3. Find the total number of bound states N of the bounded potential

$$V(x) = -u(x),$$

   where $u(x) > 0$ with $u(x) \to 0$ as $x \to \pm\infty$.

4. For the non-classical region we have $E < V(x)$, and thus $p(x) = \sqrt{2m[E - V(x)]}$ is imaginary. For this, calculate the probability of tunneling of an α particle through the nuclear potential barrier.